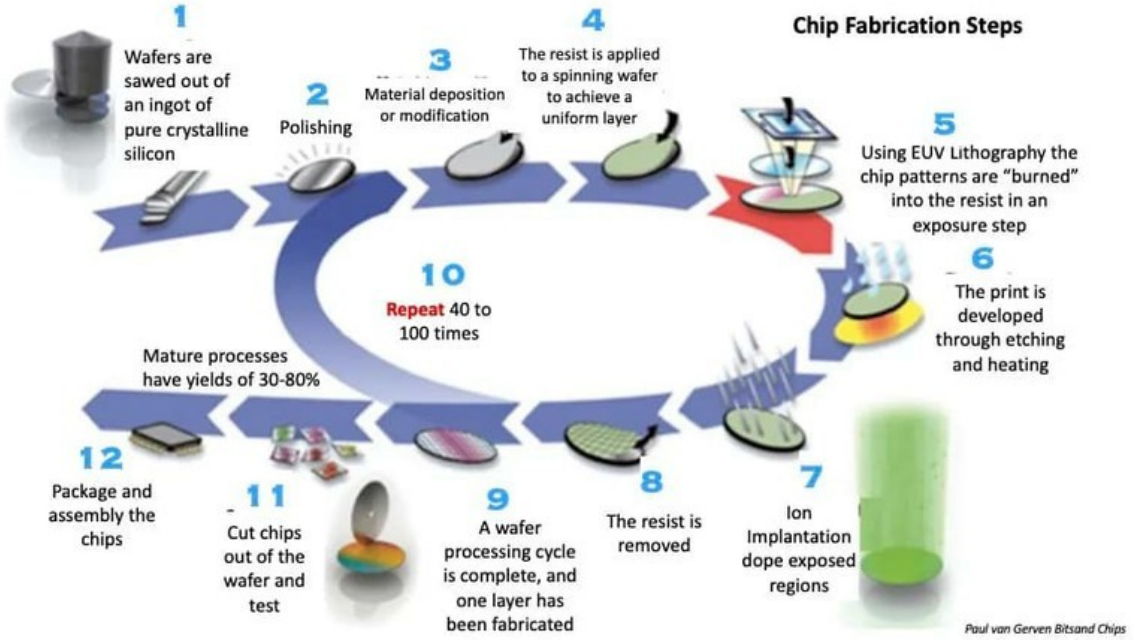
FUTUREWEI
Technologies

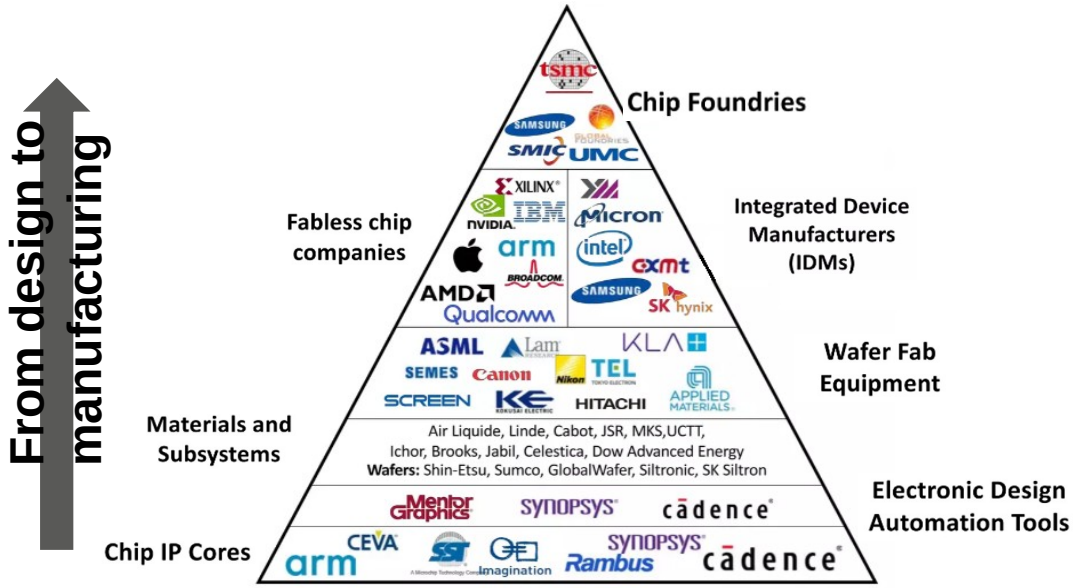
# Semiconductor Ecosystem Fabrication Process/Industry Dynamic

Ecosystem Team | Ziqi Wang

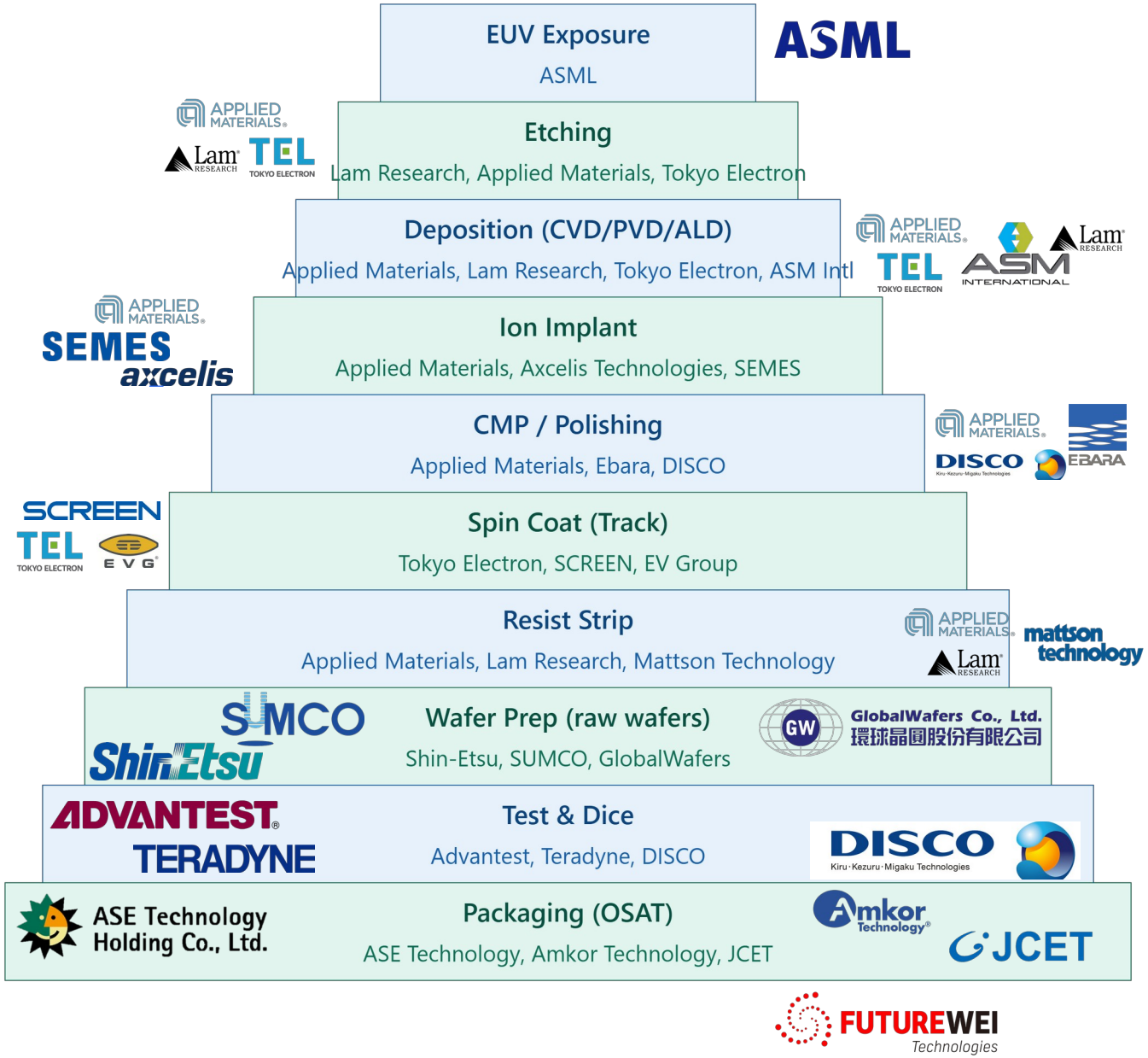
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# Chip Fabrication Process

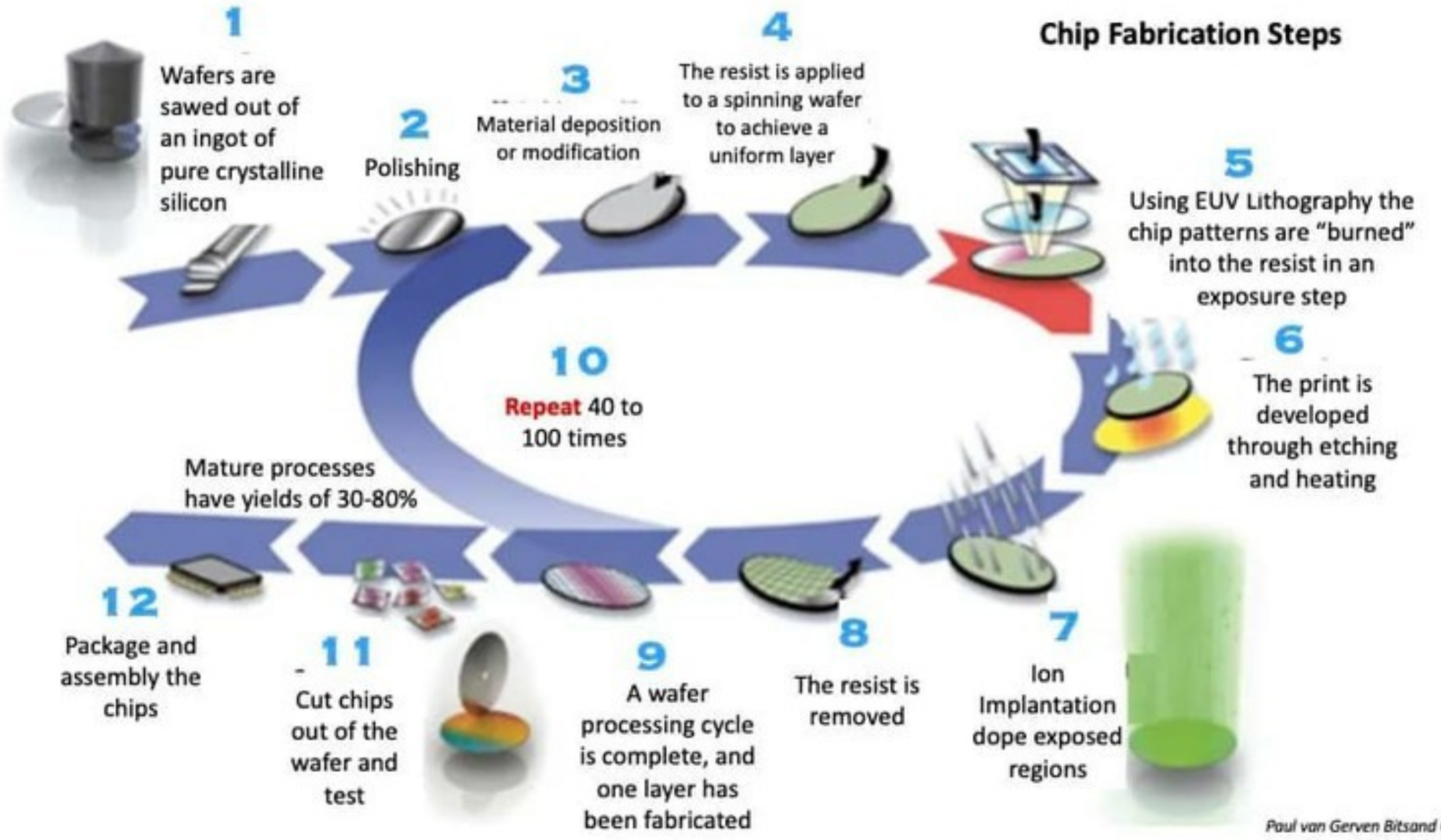



# Wafer Fab Equipment Providers




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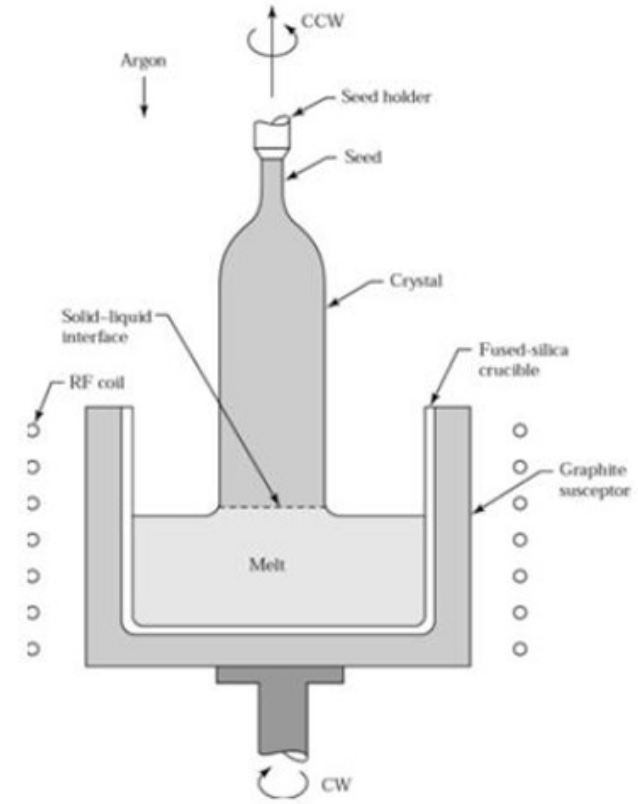

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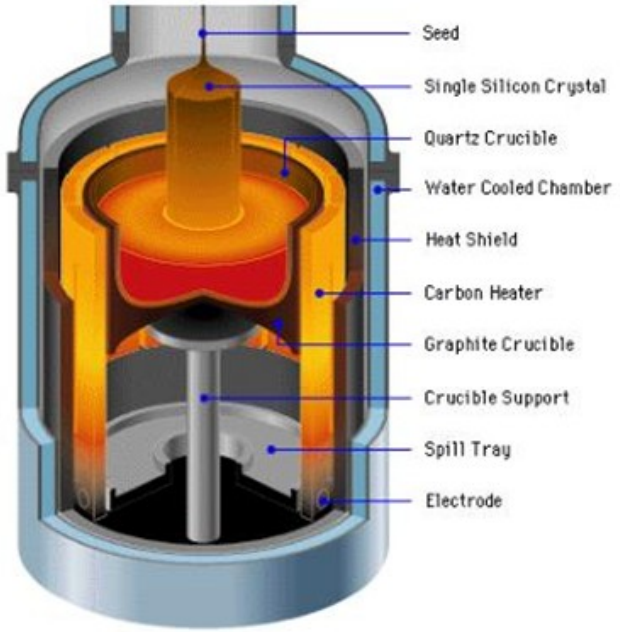
# Ingot Making Process




**Raw Material → Purify → Grow Crystal (Czochralski) → Slice & Polish**

## Czochralski method (CZ)



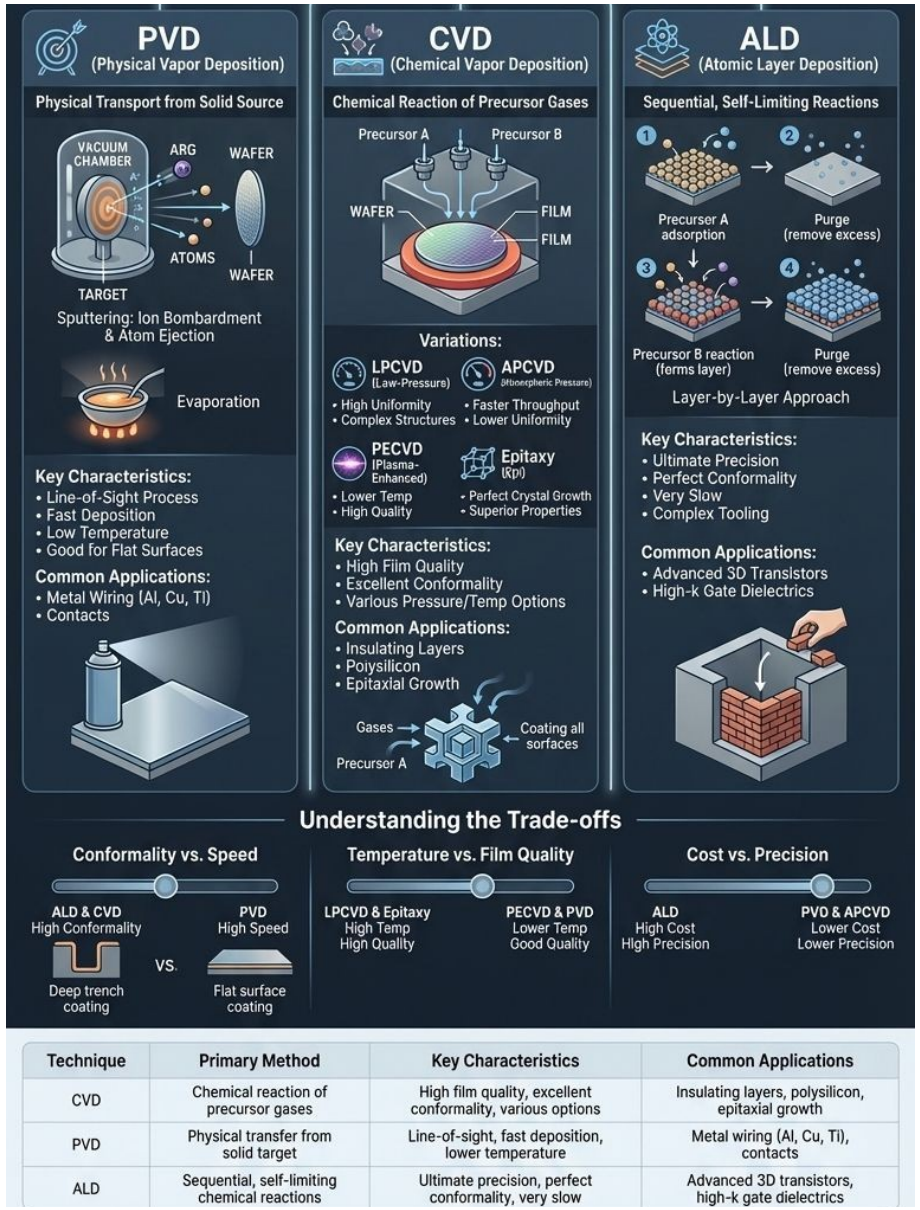



Wafer/ingot producers: Shin-Etsu Chemical (Shin-Etsu Handotai), SUMCO Corporation, GlobalWafers, Siltronic AG, SK Siltron

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# Deposition



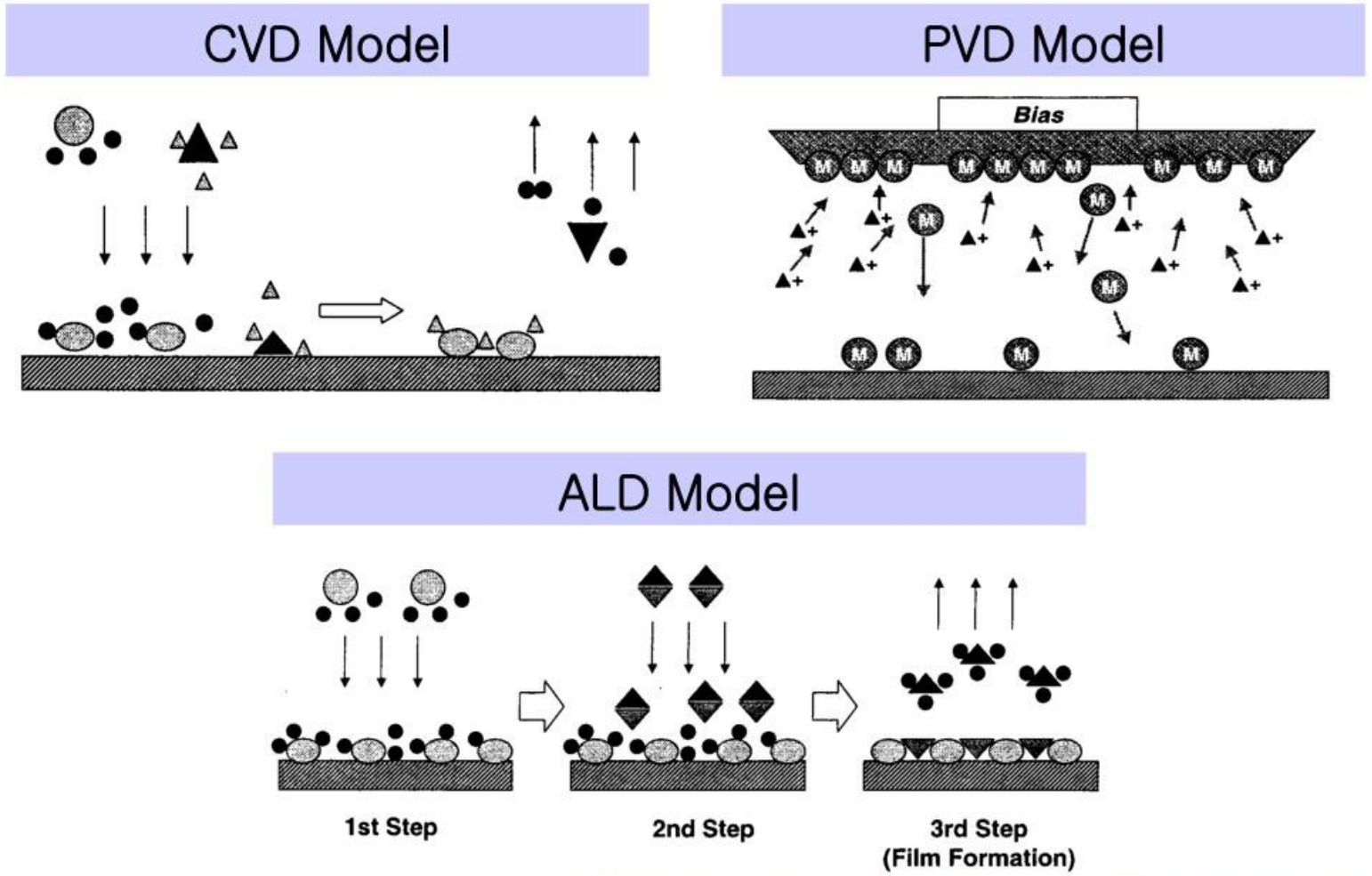




| Technique | Primary Method | Key Characteristics | Common Applications |
|---|---|---|---|
| CVD | Chemical reaction of precursor gases | High film quality, excellent conformality, various options | Insulating layers, polysilicon, epitaxial growth |
| PVD | Physical transfer from solid target | Line-of-sight, fast deposition, lower temperature | Metal wiring (Al, Cu, Ti), contacts |
| ALD | Sequential, self-limiting chemical reactions | Ultimate precision, perfect conformality, very slow | Advanced 3D transistors, high-k gate dielectrics |

- CVD
  - Reactive gases (precursors) are introduced into a chamber, where they react near or on a heated wafer to form a solid film.
- PVD
  - Material is vaporized from a solid source (target) and deposited onto the wafer without chemical reaction.
- ALD
  - Sequential, self-limiting chemical reactions deposit ultra-thin, conformal films even on complex topographies.



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CVD is the most widely used by volume, PVD dominates metal layers, ALD is the smallest but the one every advanced-node roadmap is moving toward.

**CVD providers**: Applied Materials, Lam Research, Tokyo Electron, ASM International, Kokusai Electric
**PVD providers**: Applied Materials, Tokyo Electron, ULVAC, Veeco Instruments, AJA International
**ALD providers**: ASM International, Tokyo Electron, Applied Materials, Lam Research, Aixtron

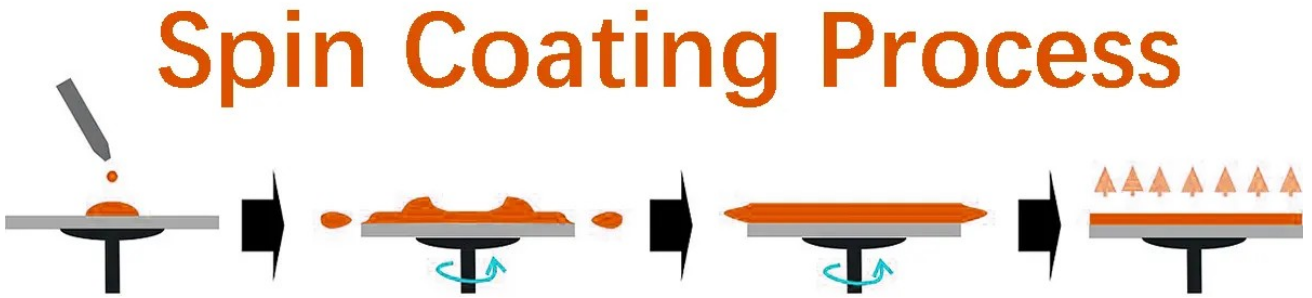


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# Spin Coating



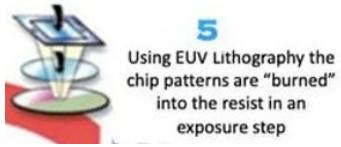

- **Dispense:** Liquid precursor is dropped onto the center of the stationary or slow spinning wafer
- **Spin Up:** Wafer accelerates to target speed; centrifugal force spreads the liquid outward and flings excess off the edge
- **Fluid Outflow:** At steady rotation speed, viscous flow continues thinning the film into a smooth, uniform layer
- **Evaporation:** Solvent evaporates from the thin film, leaving a solid coating on the wafer surface
- **Soft Bake:** Wafer is heated on a hotplate to fully evaporate residual solvent, stabilizing the film before exposure



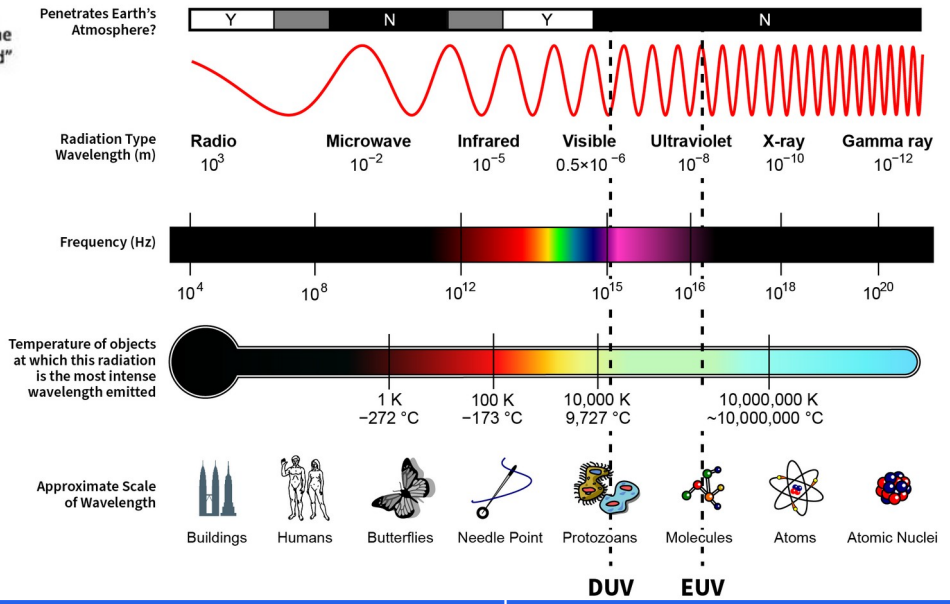



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# Exposure (lithography)




- Electromagnetic radiation is projected through a **photomask** or **reticle** containing the desired circuit pattern.
- The radiation alters the chemical structure of the photoresist in the exposed areas.
- Shorter wavelengths enable finer pattern resolution, which is why the industry moved from DUV to EUV.

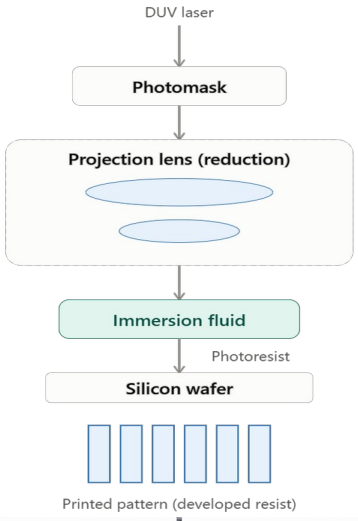


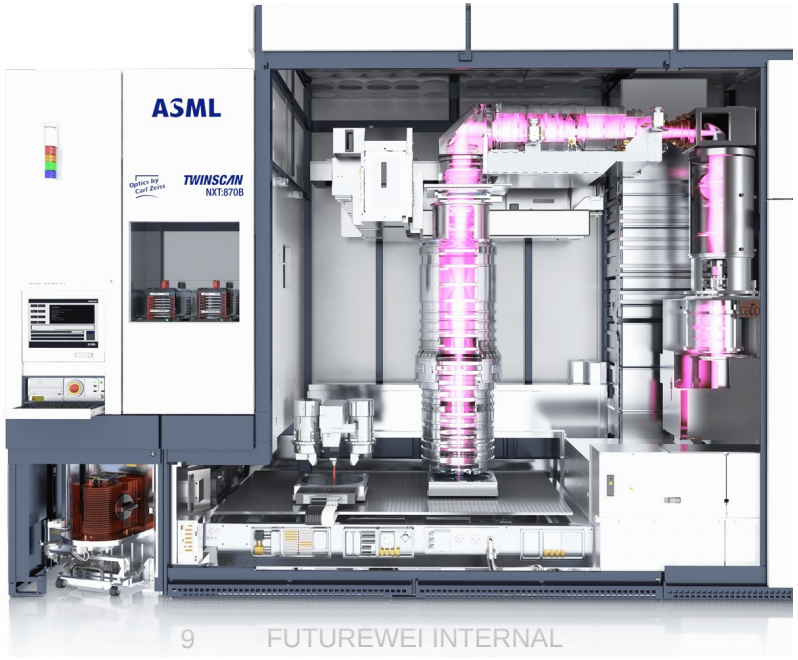

| | DUV Immersion | EUV | High-NA EUV |
|---|---|---|---|
| **Wavelength λ** | 193 nm | 13.5 nm | 13.5 nm |
| **Numerical Aperture** | 1.35 | 0.33 | 0.55 |
| **Target Node** | 65–22 nm | ≤ 7 nm | ≤ 2 nm |
| **Status** | Mature | HVM today | From 2025 |
| **Providers** | ASML, Nikon, Canon, SMEE (Shanghai Micro Electronics Equipment) | ASML | ASML |

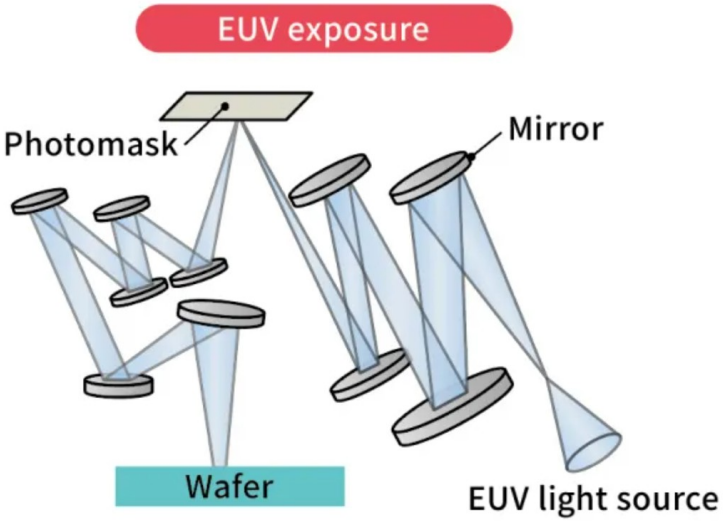


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# DUV

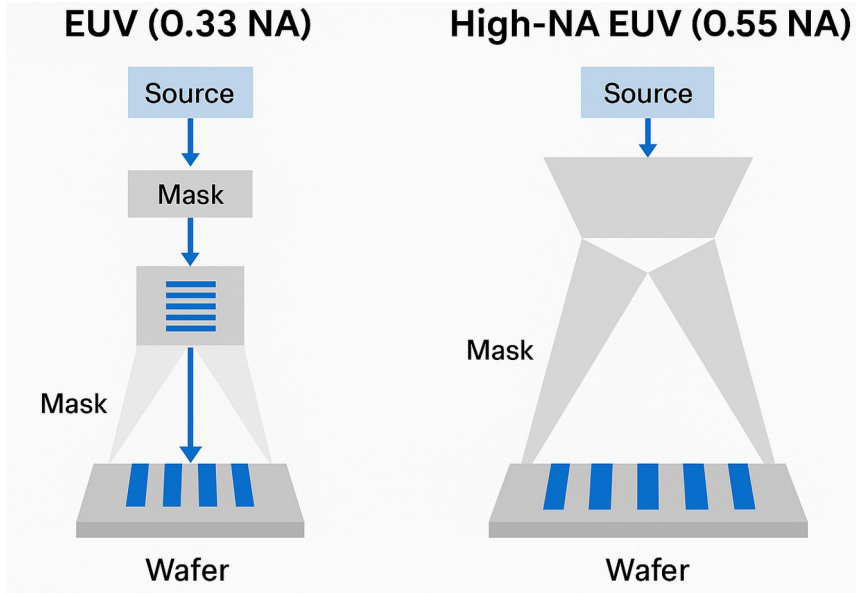


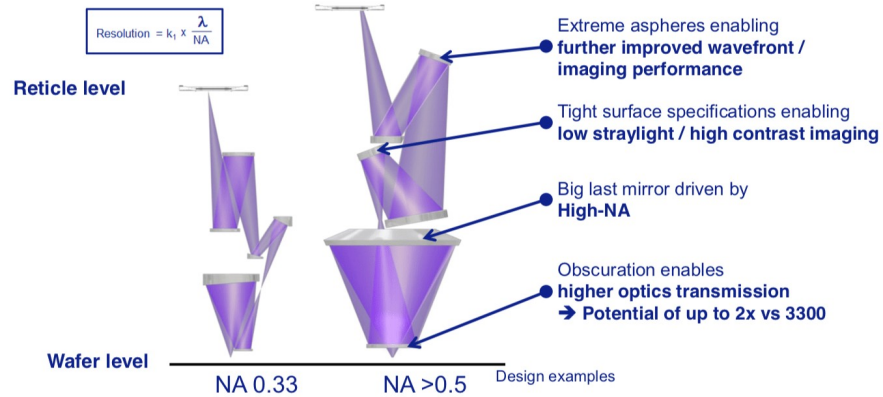






# High-NA EUV



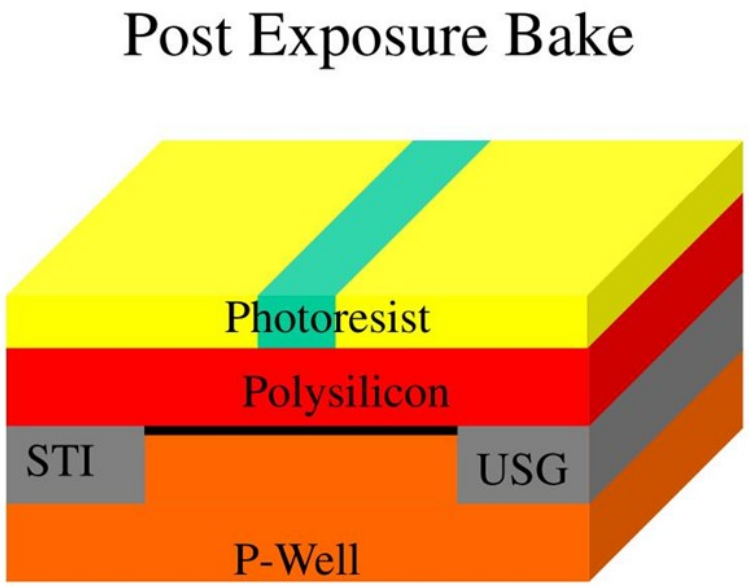



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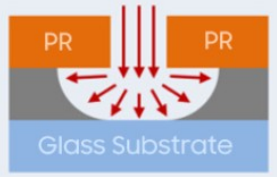
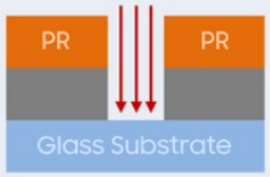
# Post Exposure Bake & Etching




**Stabilizes the resist profile; Enhances chemical reactions; Improves solubility contrast; Promotes cross-linking**




## Comparison of Two Etching Methods

| | Wet Etching | Dry Etching |
|---|---|---|
| Method | Chemical Reaction (Solution) | Physical/Chemical Reaction (Gas) |
| Pros | Low Cost<br>High Speed<br>Simple Process | High Precision<br>Enables Micro-Patterning |
| Cons | Low precision<br>Chemical Contamination<br>Etching Beneath PR | High Cost<br>Low Speed<br>Complex Process |
| Etching Direction |  |  |





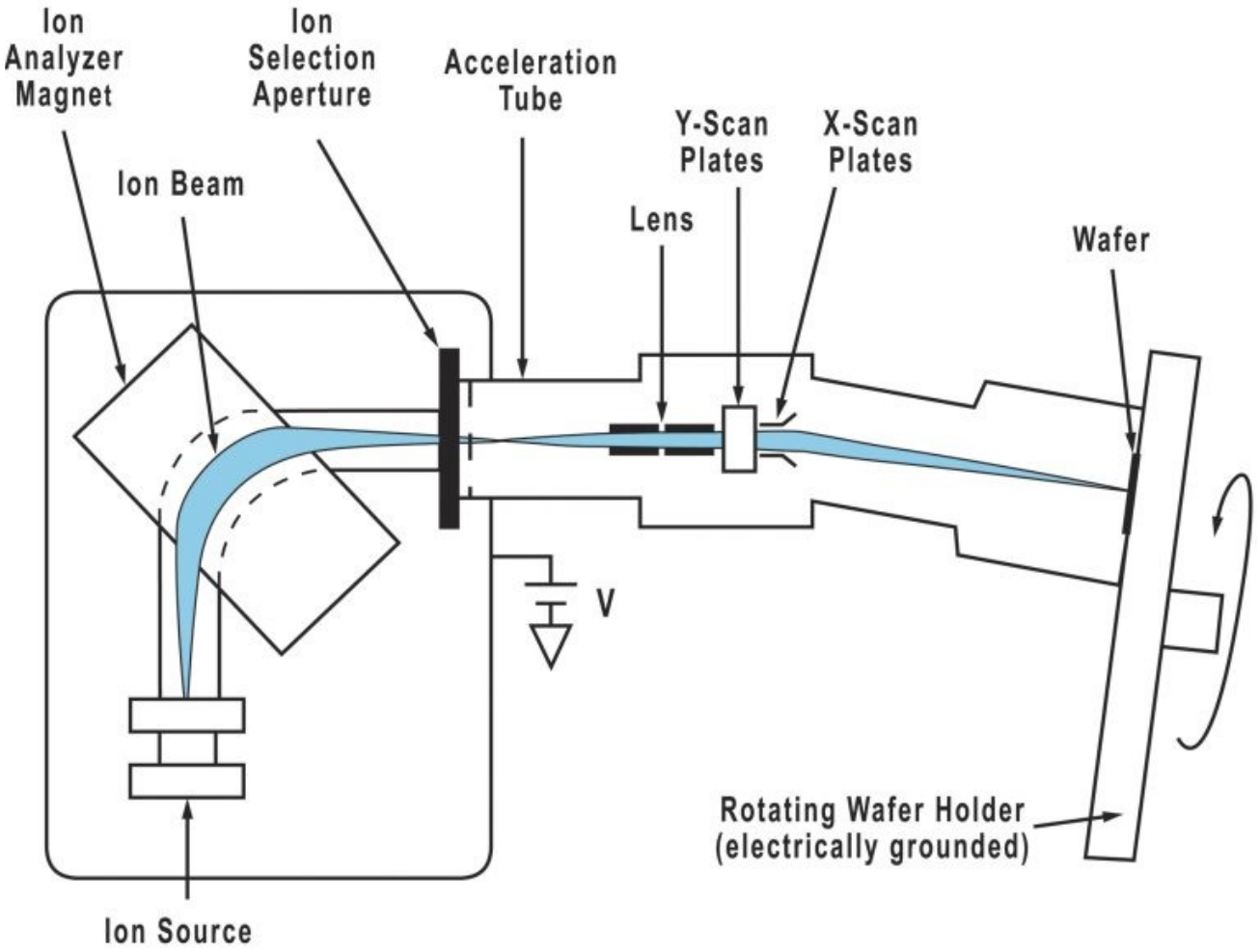


Providers: Lam Research, Applied Materials, Tokyo Electron, Hitachi High-Tech, NAURA

# Ion Implantation




- Introduces dopant atoms into silicon to create p-type and n-type regions
- Preferred over other doping methods because it delivers precise doping with extremely pure dopant species, ideal for fine geometries
- Process flow: ion source generates plasma → mass analyzer isolates the target ion species → acceleration column brings ions to the required energy → lens and scan system sweep the beam evenly across the wafer
- **Dose**: number of ions per unit area, controls doping concentration
- **Acceleration Energy**: controls how deep ions penetrate into the wafer
- **Beam Current**: higher current speeds up doping but risks wafer overheating
- **Implantation Angle**: tilted about 7 degrees off normal to prevent channeling

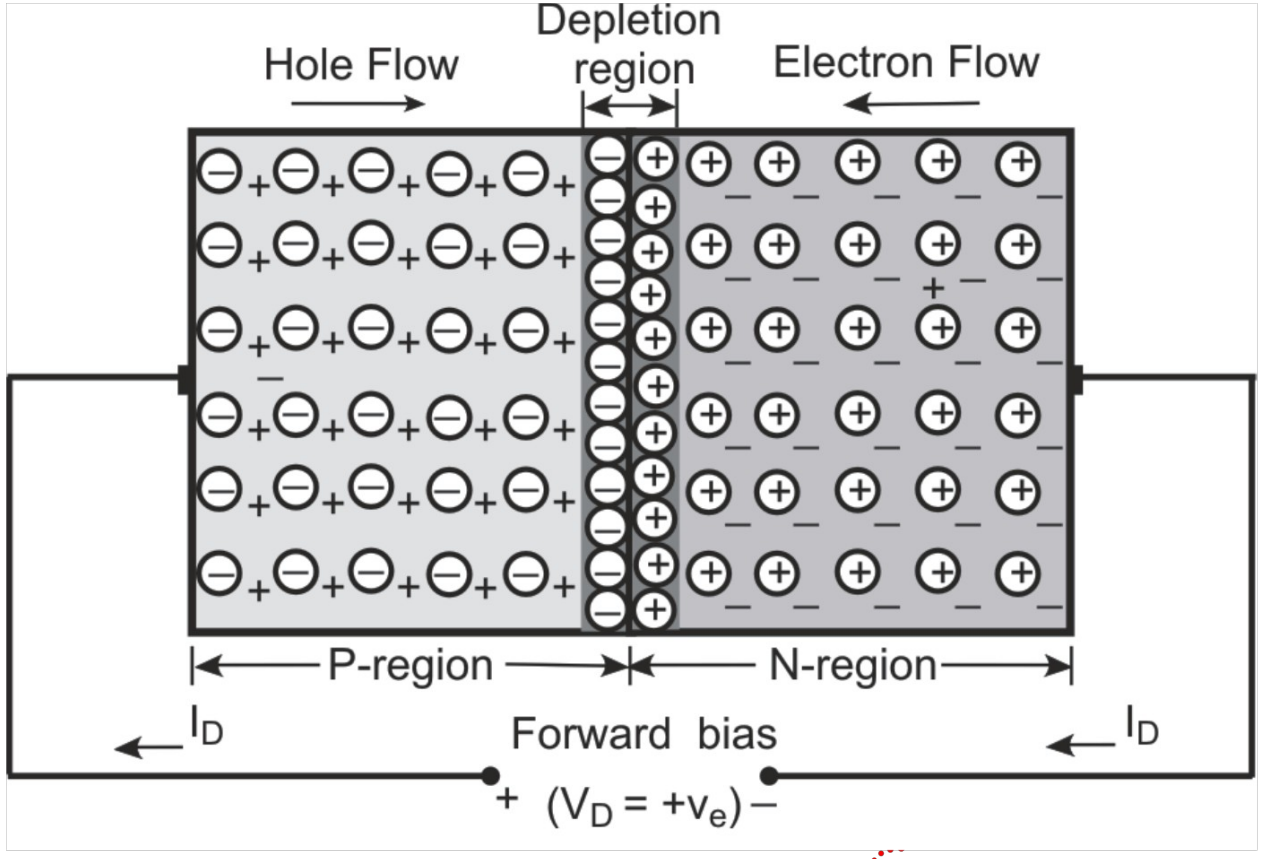





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# PN Junction and Depletion Region

- P type and N type regions form a depletion region at the junction
- Depletion region blocks free carrier flow at equilibrium
- Forward bias narrows the depletion region
- Holes and electrons cross the junction and sustain current I_D
- This is the basis for source and drain junctions in a MOSFET

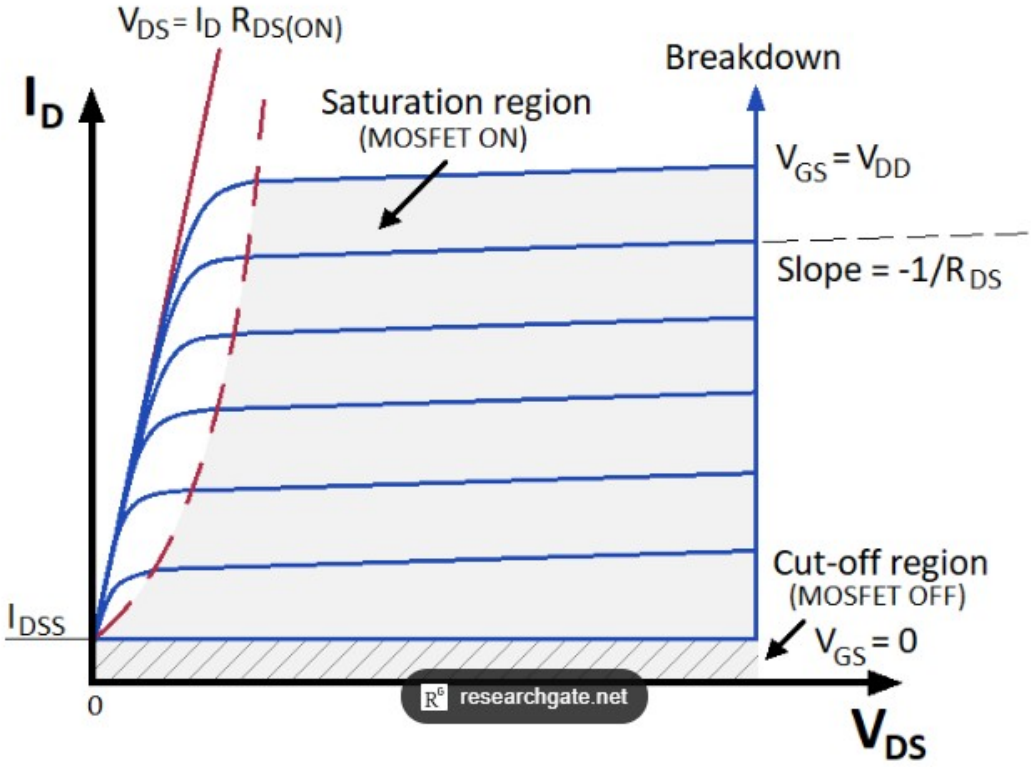





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# MOSFET Output Characteristics

- V_GS = 0 places the device in cut off
- Cut off corresponds to the switch OFF state
- V_GS near V_DD places the device in saturation
- Saturation current levels off and corresponds to switch ON
- Gate voltage controls the transition between these states

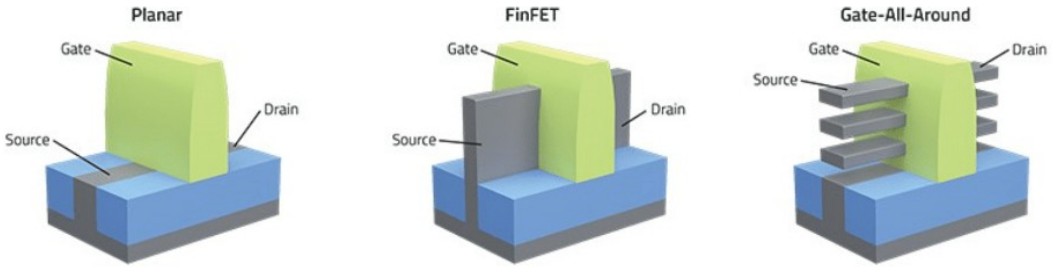





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# Scaling Beyond FinFET: GAA Transistors and HBM Memory

## GAA (Gate-All-Around)




**GAA providers (foundries):** Samsung (MBCFET, since 2022), TSMC (N2 nanosheet), Intel (RibbonFET, 18A)

## HBM Bottleneck

**HBM Technology Roadmap: From HBM3 to HBM4**

| Product | HBM3 | | HBM3e | | HBM4 | |
|---|---|---|---|---|---|---|
| Release Time | 2022 | | 2024 | | 2026F | |
| Core Die Density | 16Gb | | 24Gb | | 24Gb | |
| Layer | 8,12 | | 8,12,16 | | 12,16 | |
| Speed (Gbps) | 5.6-6.4 | | 8.0-9.8 | | 8-10 | |
| I/O | 1024 | | 1024 | | 2048 | |
| Core DRAM Node | 1z | SK hynix / SAMSUNG | 1a / 1b | SAMSUNG / SK hynix micron | 1c / 1b | SAMSUNG / SK hynix micron |
| AI Accelerator Clients/ Products | NVIDIA / AMD | Hopper / Instinct MI300X | NVIDIA / AMD | Blackwell / Instinct MI350X | NVIDIA / AMD | Rubin / Instinct MI400X |

Source: TrendForce, company press releases

NEWS

**HBM providers (memory makers):** SK hynix, Samsung, Micron



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# Roadmap in Packaging




## From 1D to 3D Semiconductor Packaging

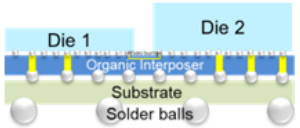
IDTechEx

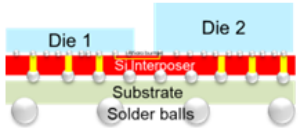
| 1D | 2D | 2D enhanced (2.1D – 2.5D) | | 3D |
|---|---|---|---|---|
| Board level integration | Package level integration | Wafer level integration | | |
| | | Organic interposer | Si interposer | |
| *Different packaged chips assembled on a PCB board* | *Integration of different packages/components on a unifying substrate* | *Integration of different dies on an organic interposer/redistribution layer (RDL) and packaged on a unifying substrate* | *Integration of different dies on a Si interposer or Si bridge and packaged on a unifying substrate* | *Integration of different dies on top of each other and packaged on a unifying substrate* |

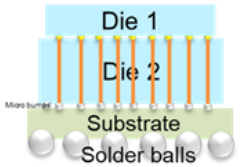



***Multi-chip Module (MCM)***



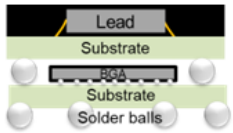




IDTechEx Research

***Package on package (PoP)***

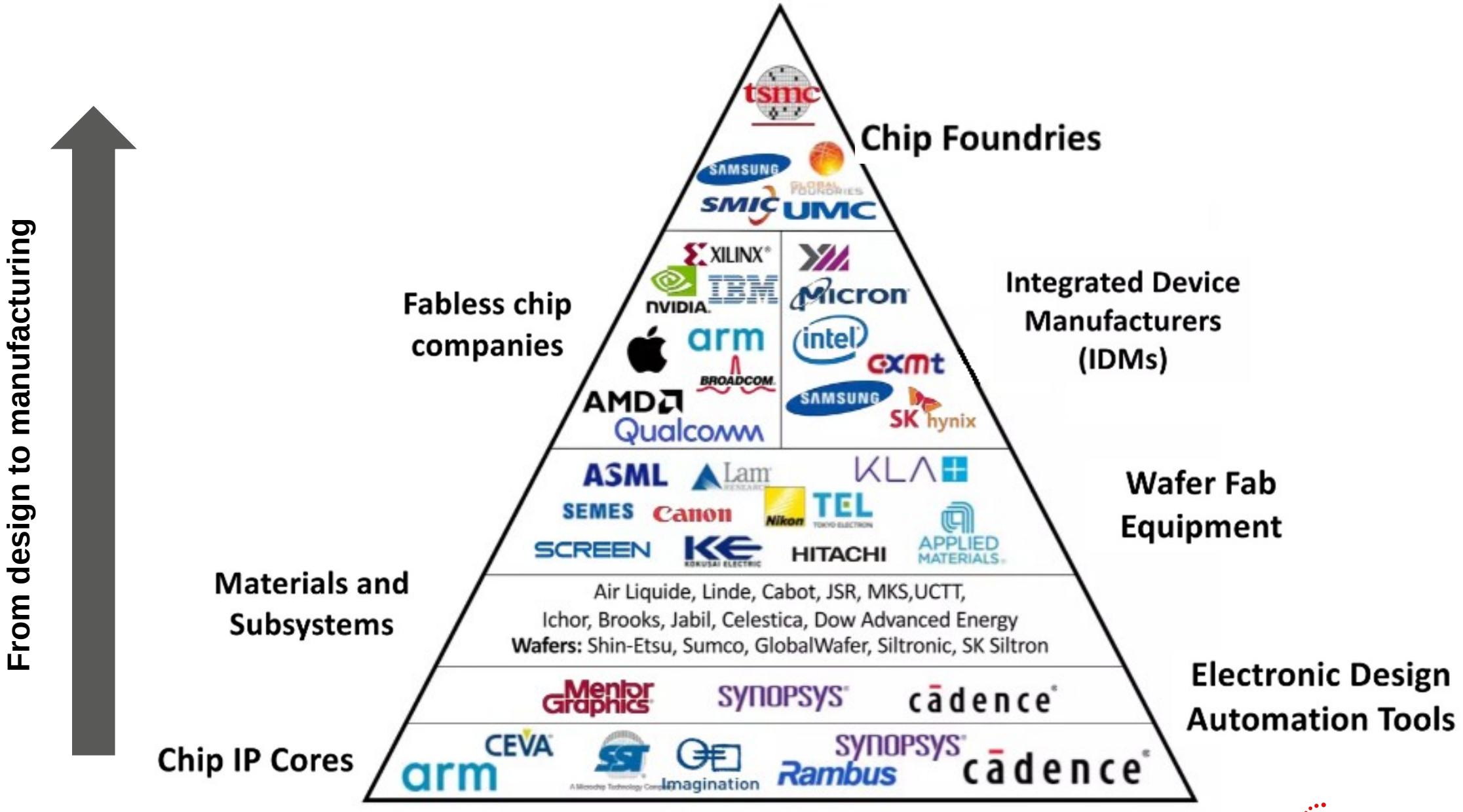



Low → High

**Interconnect density, Level of application,**
**Cost, Process challenges**



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# Thank You.

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